

Data Sheet Pick-by-Light

SAP number Pick-by-Light: 10568231

Prerequisite/requirement to use the option

Clarify the status of the Sutwin software (only applies to retrofits, since new machines are equipped with the latest software). Starting with Sutwin version 5. the option should run without problems, otherwise consult development.

Important to know

The "Pick-by-light" option has no wear parts and therefore no maintenance costs.

Compatible with all Splicer machines with RS232 connection.

Voraussetzung/Anforderung zum Benutzen der Option

Function

The pick-by-light system is designed to help the machine operator find and pick of the wires to be welded facilitate and speed up the process. For this LED's above the lines for the next welded connection are required are switched on automatically. The other LEDs are corresponding switched off. The output channels of the pick-by-light hardware are connected to LEDs that be attached to a rack with various lines to be gripped.

Up to 32 LEDs can be connected to a pick-by-light hardware.

It is possible to cascade up to eight devices, so that in total up to 256 LEDs can be connected to a machine.



Additional function

One option is to turn off the LEDs after grabbing the line.

It can be controlled by any sensors (buttons, motion sensor, etc.). (Sensors, buttons must be obtained and installed by the customer himself). It is possible to trigger the welding only when all the wires removed by the operator and the associated sensors/buttons have been activated. This becomes visible when the respective LED goes out when the button is pressed.

Example:



The green LED shows which line is being used and is then confirmed by pressing the button.



After actuation, the green LED goes out and signals the release "Welding released". Only now can the welding be triggered.

Installation

For the basic function of the system, the pick-by-light hardware is used via RS232 connected to the welding machine. After activating the system and setting up the desired options (Switch off the LEDs / deactivate the welding process) is the assignment of the lines to the corresponding LEDs. Pick-By-Light is only available from the Pick-by-Light PLC software version "Pick by Light 2.0" available and therefore does not work with older versions.

PickByLight is only designed for integration into Sutwin. Sutwin provides all the necessary conditions available.

The welding parameters, the behavior of the machine and the quality of the product are not affected by this feature.



software installation

A new [PickByLight] section must first be created in the sut.ini file
become.

Now in the sut.ini section [PickByLight] the number of the LED (output port
the pick-by-light hardware) is assigned to the name of the line as follows
become:

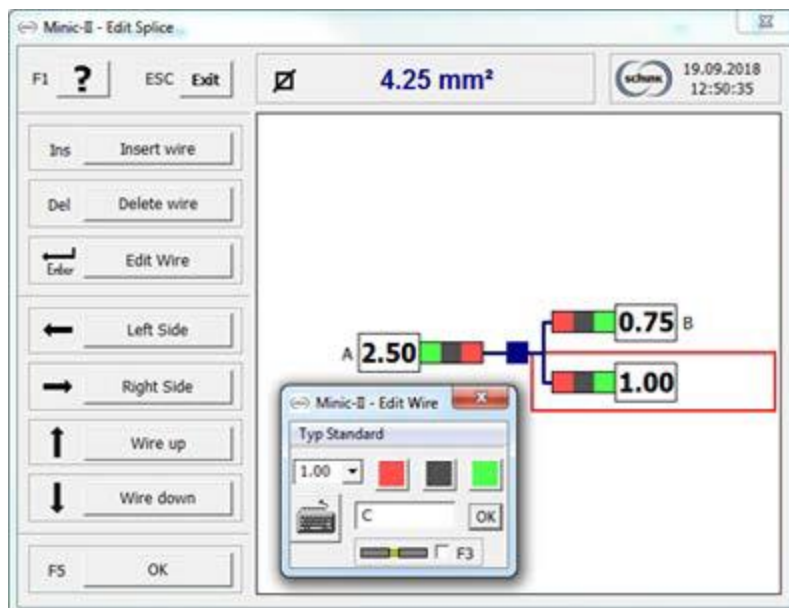
[PickByLight]

0,75mm²=1

1,0mm²=2

2,5mm²=3

usw.



The "WIRENAME.." entries must match the names assigned in the splice editor
to match.

The following should be entered as an example for the wires specified in Fig. 4
become:

[PickByLight]

A=1

B=2

C=3



When equipping the containers for the lines, these settings must now be made
are taken into account. According to the example above
the lines with a cross-section of 2.5 mm² (A) are in the container
with the LED connected to output port 1 of the pick-by-light hardware is.
Save the settings in the sut.ini file and exit the text editor to
restart the Sutwin software

scope of delivery

The option includes the following hardware (picture)
+ supply line to a 230V network and an RS232 cable to
connect to the PC of the respective machine.



customer installation

The customer has to get a frame as shown in the picture below, as well as the buttons and sensors.

In other words, the frame, sensors and buttons are not included in the scope of delivery.



Quickstep instructions for activation of the option

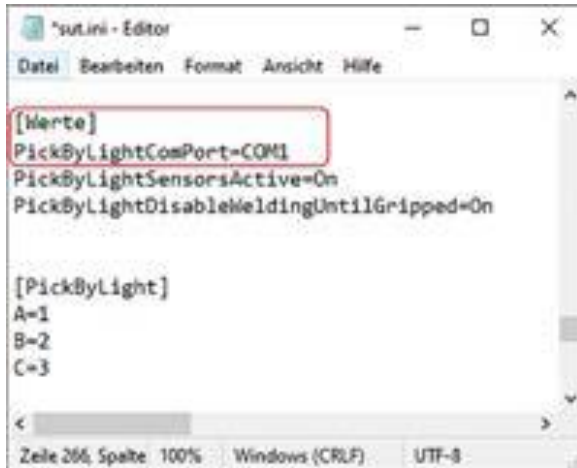
1. Connect LEDs and sensors to PbL box (not made by Schunk)
2. Connect the PbL to the machine via the RS232 connector
3. Power PbL by plug
4. Turn on PbL
5. Switch on the US welding machine
6. Activate PbL in Sut.ini
 - a. Key position on the right - setup mode / set up nodes
 - b. Press ESC or "Exit"
 - c. Query "Shut down Windows?" answer with No / ESC
 - d. Enter password "1902".
 - e. Open C:\Sut



f. Open the "sut" or "sut.ini" file

g. Text editor opens

h. Enter PickByLightComPort=COM1 or =COM2 under [Values] (depending on which port PbL was plugged into)



7. Then assign the cable label to the PbL connectors

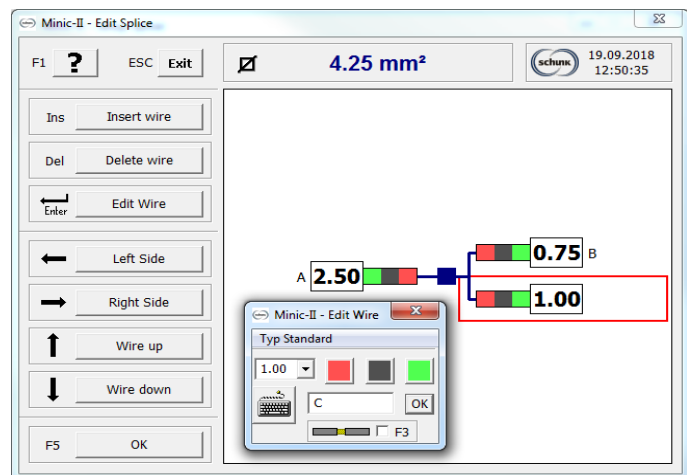
a. For this, the "[PickByLight]" section must be entered manually

b. Then allocation such as "A=1" or "SchunkKabel6sqmm=1" or "075sqmm=1"

c. A = name of cable in node

d. 1 = port to PbL

e. Example: here A and B (picture)



8. File + Save + close the text editor or close the text editor and answer the Windows query with "Save".

9. start sutwin again by double-clicking (sutwin app / sutwin.exe)





Sonosystems®

